

# Exercise - 9 Solutions

## Out-of-Distribution Detection

### Exercise 9.1: The OOD Problem

Your CARLA model was trained on sunny daytime images. At deployment, it may encounter images from fog or night - conditions outside its ODD.

1. Why can a standard classifier not be trusted to signal when it receives an OOD input?

→

A standard classifier cannot be trusted to signal OOD inputs because it is trained only to classify inputs into its known classes. It does not learn a separate “unknown” or “outside ODD” class.

Therefore, when it receives foggy or nighttime images, it still produces a normal prediction and confidence score. This is unsafe because standard neural networks can assign high confidence even to OOD inputs.

2. Why is silent failure (confidently wrong prediction) worse than uncertain failure in a safety-critical system?

→

Silent failure is worse than uncertain failure because the system may continue operating normally while being wrong.

For example, if a foggy image contains a pedestrian but the model confidently predicts “no pedestrian,” the planner may not brake.

In contrast, if the model is uncertain, the system can slow down, request human takeover, or switch to a safe fallback mode.

Therefore, in safety-critical systems, detecting uncertainty or OOD inputs is essential to avoid confident but unsafe decisions.

## Exercise 9.2: Baseline OOD Detection

Describe the maximum softmax probability (MSP) baseline for OOD detection. How does it work, and what are its main limitations?

→

Maximum Softmax Probability - It is a simple baseline method for detecting out-of-distribution inputs using the classifier's own confidence.

For a classifier, the model outputs probabilities for each known class:  $p(y|x)$

$$MSP(x) = \max p(y|x)$$

For OOD detection - the outlier score is defined as

$$D(x) = - \max p(y|x)$$

| Case                 | Maximum softmax probability | OOD score      |
|----------------------|-----------------------------|----------------|
| Confident prediction | High                        | Low OOD score  |
| Uncertain prediction | Low                         | High OOD score |

Then we choose a threshold:  $D(x) > \tau$

Limitations → usually only works to some degree.

| Limitation                                 | Explanation                                                                                       |
|--------------------------------------------|---------------------------------------------------------------------------------------------------|
| OOD inputs can still get high confidence   | May confidently classify foggy, nighttime, or unknown images.                                     |
| Softmax confidence is not true uncertainty | A high probability does not prove that the input is in-distribution.                              |
| Depends on threshold                       | Choosing the right threshold is difficult and creates a false positive / false negative tradeoff. |
| Only uses final probabilities              | MSP ignores deeper feature information or whether the input actually looks like training data.    |

|                   |                                                                                  |
|-------------------|----------------------------------------------------------------------------------|
| Can fail silently | If an OOD image receives high confidence, the system may not trigger a fallback. |
|-------------------|----------------------------------------------------------------------------------|

## Exercise 9.3: Alternative Methods

Describe one more advanced OOD detection method (e.g., energy score or Mahalanobis distance). How does it improve over MSP?

→

**Energy Score** (Logit based) →

Energy score uses the model's **raw logits** before softmax. Softmax turns these logits into probabilities, but this conversion can make the model look very confident even for OOD inputs.

$$D(\mathbf{x}) = -\log \sum_i^K e^{f_i(\mathbf{x})}$$

| Energy value | Interpretation                                    |
|--------------|---------------------------------------------------|
| Low energy   | Input likely belongs to the training distribution |
| High energy  | Input may be out-of-distribution                  |

$D(x) > \tau \rightarrow$  If the energy score is above the threshold, the image is treated as OOD.

→ Energy score improves over MSP because -

it uses the **full logit information** (Considers all class logits together) instead of only the maximum probability after softmax.

**Mahalanobis Distance Method** (Feature Based) →

This method looks at the model's **internal feature representation**.

Idea – When an image input passes through a NN, the model creates feature vectors inside the network. Is this feature vector close to the normal training features, or is it far away?

If it is close to the training distribution, it is likely **in-distribution**.  
If it is far away, it may be **out-of-distribution**.

Plot a Gaussian distribution with mean and covariance to in-distribution features, then using Mahalanobis distance to measure how likely a point is under that Gaussian

$$D(x) = \sqrt{(x - \mu_k)^\top \Sigma^{-1} (x - \mu_k)}$$

| Distance       | Meaning                                                            |
|----------------|--------------------------------------------------------------------|
| Small distance | Feature vector is close to normal training features → likely ID    |
| Large distance | Feature vector is far from normal training features → possible OOD |

→ Mahalanobis Distance method improves over MSP because -  
by checking whether the internal features look similar to the training data but MSP only uses the final confidence score

**Note** - Mahalanobis does not compare the image to every training image. It compares it to the **estimated distribution** of training features (complete class) unlike KNN - directly compares against stored training examples.

## Exercise 9.4: Visualising the Distribution Shift

1. Display five in-distribution (sunny / daytime) images next to five fog and five night images.

→ (ref ipynb for implementation)

2. Also display five images from the different CARLA town. How do they differ from your training images, and how does that difference compare to the fog / night shift?

→ (ref ipynb for implementation)

→ The different-town images are visually closer to the training images than fog or night images because they still have sunny daytime lighting and good visibility. However, they differ in the structure of the environment, such as road geometry, building layout, vegetation, sidewalks, and background objects.

Compared to the fog and night shifts, the different-town shift is more subtle. Fog changes visibility and contrast, while night changes the entire lighting distribution (strong visual shifts)

The different-town images do not strongly change weather or lighting, but they change the scene layout and background structure. Therefore, different-town images are still a distribution shift, but they are less visually obvious than fog and night.

3. Compute the mean softmax confidence of each of your three models on in-distribution vs. fog/night images. Is the model more or less confident on these inputs?

→ (ref ipynb for implementation)

| Condition       | Pedestrian Detector | Traffic Light Detector | Vehicle Detector |
|-----------------|---------------------|------------------------|------------------|
| In-distribution | 0.9627              | 0.9884                 | 0.9633           |
| Fog             | 0.9280              | 0.8635                 | 0.9078           |
| Night           | 0.8280              | 0.9434                 | 0.8548           |

→

| Model                  | Confidence change                                                                     |
|------------------------|---------------------------------------------------------------------------------------|
| Pedestrian Detector    | Confidence drops from 0.9627 to 0.9280 in fog and 0.8280 at night.                    |
| Traffic Light Detector | Confidence drops from 0.9884 to 0.8635 in fog, but remains high at night with 0.9434. |
| Vehicle Detector       | Confidence drops from 0.9633 to 0.9078 in fog and 0.8548 at night.                    |

Observation → Even though confidence decreases on OOD data, it is still quite high. Models are still making confident predictions on images outside their training distribution.

## Exercise 9.5: Is the Different Town Out-of-Distribution?

The different-town images were taken under nominal weather and lighting - the kind of conditions your system is meant to handle - but in a location your models never saw during training.

1. Look back at the ODD you specified in Exercise 2.2. As written, does your ODD clearly decide whether the different-town images are inside or outside it?

→

- The ODD does not clearly decide whether the different-town images are inside or outside it. The images satisfy the weather, lighting, and camera conditions because they are sunny daytime images with clear visibility and a normal forward-facing camera.
- They also appear to satisfy the scene-type condition because they show urban roads with structured lanes.
- However, the ODD does not explicitly mention whether unseen towns, new road geometries, different building layouts, vegetation, sidewalks, or unfamiliar intersections are allowed.
- Therefore, the different-town images are not clearly outside the ODD like fog or night, but they represent an ambiguous distribution shift.

2. Revise the wording of your ODD so that the new town is unambiguously either inside or outside it. State which you chose and why.

→

Consider it **outside the ODD**.

Revised ODD-

| Dimension | Operating / valid                  | Non-operating / invalid             | Runtime detection                                             |
|-----------|------------------------------------|-------------------------------------|---------------------------------------------------------------|
| Weather   | Clear, cloudy, no precipitation    | Rain, fog, snow, hail               | Image brightness variance drops; dedicated weather classifier |
| Lighting  | Daytime with normal sunlight       | Night, dusk/dawn, tunnels, glare    | Mean pixel luminance below threshold                          |
| Camera    | Forward-facing, clean, fixed mount | Obstructed, dirty, tilted, wet lens | Blur score / sharpness metric                                 |

|                       |                                                                                                            |                                                                           |                                         |
|-----------------------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------|
| Scene type / location | Urban roads with structured lanes only in validated/geofenced CARLA towns used during training and testing | Unseen towns, unseen road layouts, highways, off-road, construction zones | GPS geofence + lane structure detection |
| Vehicle speed         | Low speed / intersection approach, below approx. 50 km/h                                                   | High-speed driving                                                        | Onboard speedometer reading             |

→ Choosing this because the different towns may still have sunny daytime weather and structured roads, but its layout was never seen during training. Since road geometry, buildings, vegetation, and intersections can influence the model, the new town represents an unvalidated distribution shift.

In a safety-critical system, it is safer to treat the new town as outside the ODD until the model is tested and validated there.

3. What does that choice imply for how the town images should be treated - as inputs the system must handle correctly, or as inputs an OOD monitor should flag?

→

Because the ODD is revised so that unseen towns are outside the operating domain, the different-town images should be treated as **OOD inputs**.

This means the system is not expected to handle them as normal driving inputs. Instead, an OOD monitor should flag these images as outside the validated operating conditions.

Once flagged, the system should take a safe fallback action, such as reducing speed, requesting human takeover, or disengaging autonomous mode. This is safer than assuming the model will generalize correctly to an unseen town layout.

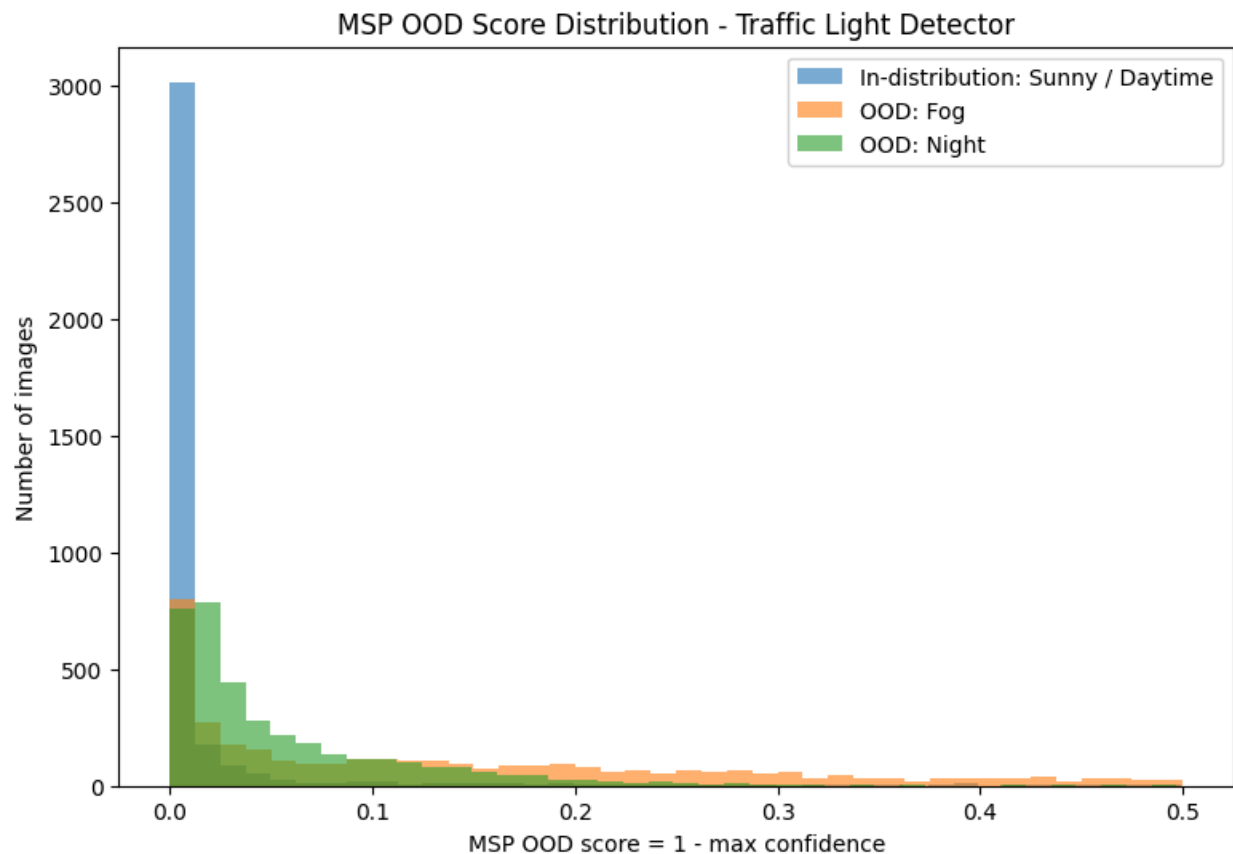
## Exercise 9.6: Evaluating the MSP Baseline

Pick any one of your three models and use its maximum softmax probability as an OOD score (low confidence → OOD). Treat the sunny/daytime test images as in-distribution and the provided OOD images as OOD.

1. Plot the distribution of OOD scores for in-distribution and OOD images.

→ (ref ipynb for implementation)

Chosen model - **Traffic Light Detector** (earlier it failed badly at night, so it is interesting for OOD detection)



i) For sunny/daytime in-distribution images, most scores are very close to 0, meaning the model is highly confident.

ii) For fog images, the distribution shifts toward higher OOD scores and becomes more spread out, showing that the model is less confident under foggy conditions.

iii) For night images, the distribution is also more spread out than the sunny distribution, but many night images still receive low OOD scores. This means the model remains confident on many night inputs even though they are outside the ODD.

Overall, MSP detects some distribution shift, especially for fog, but it is not fully reliable because the OOD and in-distribution score distributions overlap. Hence, MSP is not very useful here.

2. Compute the AUROC for separating in-distribution from OOD, over all of the OOD scenarios.

→ (ref ipynb for implementation)

AUROC - The probability that a randomly chosen OOD image receives a higher OOD score than a randomly chosen in-distribution image.

In this case **AUROC = 0.8858**

→ The AUROC for separating sunny/daytime in-distribution images from fog and night OOD images using the MSP OOD score is **0.8858**. This indicates that the MSP baseline provides reasonably good separation between ID and OOD inputs. In other words, an OOD image receives a higher OOD score than an in-distribution image in about **88.58%** of random pairs.

As per the plot, the score distributions still overlap, meaning that some fog or night images are assigned low OOD scores and are therefore treated as confident in-distribution predictions. This confirms that MSP is useful as a simple baseline, but it is not reliable enough as the only safety mechanism for OOD detection.